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Geographical Indications and their Impact on Territorial Development: Empirical Evidence for Brazilian Municipalities

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ABSTRACT

The promotion of Geographical indications (GIs) as a development intervention has gained growing support. However, despite discussions related to the development resulting from GIs, studies quantifying the impact of public policies for geographical indications are still scarce. In an attempt to fill in this gap, this paper evaluated the effect of geographical indications (GIs) in the process of territorial development of Brazilian municipalities, focusing on the coffee and wine GIs. Therefore, we employed the Propensity Score Matching method. Our results demonstrated a positive mean effect for most of the variables selected, evidencing the importance of the GIs for territorial development.

Keywords: Geographical indications; Territorial development; Coffee; Wine.

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Indicaciones Geográficas y su Impacto en el Desarrollo Territorial: Evidencias Empíricas para los Municipios Brasileños

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RESUMEN

La promoción de las indicaciones geográficas (IG) como una intervención de desarrollo ha ganado un apoyo creciente. Sin embargo, a pesar de las discusiones relacionadas con el desarrollo derivado de las IG, los estudios que cuantifiquen el impacto de las políticas públicas para las indicaciones geográficas aún son escasos. En un intento por llenar este vacío, este trabajo evaluó el efecto de las indicaciones geográficas (IGs) en el proceso de desarrollo territorial de los municipios brasileños, enfocándose en las IG de café y vino. Por lo tanto, empleamos el método *Propensity Score Matching*. Nuestros resultados demostraron un efecto medio positivo para la mayoría de las variables seleccionadas, evidenciando la importancia de las IG para el desarrollo territorial.

Palabras clave: Indicaciones geográficas; Desarrollo territorial; Café; Vino.

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1. Introduction

Geographical indications (GIs) identify products that present a specific quality (or notoriety) associated to their geographical origin. They result from technical, social and economic interactions, which include both the mobilization of specific local resources – such as the local know-how and cultural traditions – , and the construction of the product reputation within the supply chain between producers and consumers (Belletti, Marescotti, & Touzard, 2017).

Therefore, having a GI guarantees protection against frauds related to the product origin (Besah-Adanu, Bosselmann, Hansted, & Kwapong, 2019; Chabrol, Mariani, & Sautier, 2017), and promotes a differentiation of this product in the market, improving, theoretically, these producers' position regarding competitiveness and profitability. Dias e Mendes (2018) point out the growing consumers' interest in the quality of traditional products from certain regions; in this sense, a GI might lead to an intensification of the demand for these goods, with the identification of their characteristics and traceability.

Especially in relation to farm products, for being homogeneous and having their price determined in an exogenous way, their differentiation might intensify the external insertion of certain region, with its valorization (Goldstein, 2001). Thus, a GI might lead from the perfect competition to a monopolistic competition by aggregating product value, with the generation of more income, and, consequently, with the possibility of greater fixation of the population in the rural area.

For this reason, joining their consequent potentials, geographical indications can be seen as a protection system (to the consumer and to the farmer); a marketing tool (emphasizing the differentials of a product or service); a development mechanism (since it might influence the generation and maintenance of jobs, income distribution, local identity, quality of life, etc.); and a preservation instrument (of the culture, *savoir-faire* and even the environment) (Medeiros, Passador, & Passador, 2016).

In this context, geographical indications might achieve several public objectives, becoming more than only an intellectual property right, also acting as a political instrument. Moreover, they receive growing support from a wide variety of stakeholders, not only the nation-states and the financing international agencies, but also from local territorial authorities, unions and NGOs, which tend to have their political agendas related to the GIs, fostering the local social capital (Arfini, Cozzi, Mancini, Ferrer-Perez, & Gil, 2019; Marie-Vivien & Biénabe, 2017).

As for Brazil, aiming at putting into practice the adhesion to international agreements regarding intellectual property[1], the country internalized the norms and enacted the new Industrial Property Law in 1996 (Law n° 9.279, LPI), with the regulation of geographical indications in articles 176 to 183 of that law (BRASIL, 1996). In addition, through the Decree 5.351, of 21st January 2005, the Departamento de Propriedade Intelectual e Tecnologia da Agropecuária - DEPTA (Farming Intellectual Property and Technology Department) was created, aiming to incentivize the geographical indication of farming products, mapping and listing the several regions and products that could be protected by GI.

In Brazil, there are two types of GIs, namely, indication of provenance (IP) corresponding to the indication of source, and denomination of origin (DO), corresponding to the appellation of origin (BRASIL, 1996; Cruz & Hespanhol, 2018). Currently, (up to the beginning of November 2020), Brazil holds 72 geographical indications, from which 58 are indications of provenance and 14 are denominations of origin. They are distributed all over the country and include products such as jaboticaba fruit, cocoa, green coffee, banana, wines and sparkling wines, cheese, shrimp, honey, propolis, cachaça (sugar cane liquor), embroidered products, stones, jewelry, among others.

[1] With the creation of the World Trade Organization (1994), discussions around intellectual property and, consequently, geographical indications were formalized. This occurred mainly with the approval of the Agreement on Aspects of Intellectual Property Rights related to trading, one of the appendices of the World Trade Organization Constitutive Act, which entered in force in January 1995.

The protection of geographical indications (GIs) worldwide has been more and more explored as a tool to support the sustainable local development (Belletti et al., 2017). More specifically, in the "global south" the GIs have gained growing political and economic relevance (Chabrol et al., 2017), as well as importance as a research theme, mainly "[... regarding the specific nature of challenges and case studies of experiences in the Global South [...]" (Egelyng et al., 2017, p.44).

Considering those, this paper aimed to evaluate the effect of geographical indications on the process of territorial development of Brazilian municipalities, more specifically, focusing on coffee and wine and analyzing their effect on total employment, tourism employment, income, exports, rural exodus and socioeconomic development. These two products present strong cultural identity[2], also, the first GIs recognized in the country were those of wine (Vale dos Vinhedos IP) and coffee (Cerrado Mineiro Region IP). In addition, the GIs strategy has been strongly adopted in the wine sector in European countries with greater tradition in this sign such as France (Diallo, 2017).

Although several relevant studies have focused on developing countries, most of the literature on GIs describes the experience in Europe, where the socioeconomic and cultural conditions are very different from those in developing countries (Mancini, 2013, p.296). In addition, the development benefits resulting from the GIs have not been confirmed yet, and empirical support and evidence of positive impacts presented from the Global South are still limited (Neilson, Wright, & Aklimawati, 2018). The national literature on the GIs impact on territorial development is limited in its selection of analysis methods, therefore, most of the studies (such as Carvalho & Dias, 2012; Nascimento et al., 2012; Silva et al., 2016) focus only on the descriptive analysis of this relation. Therefore, this paper presents a methodological advance for measuring quantitatively the GIs impact on territorial development and controlling other factors that might interfere in this process, through the Propensity Score Matching method.

This paper is organized in five sections that include this introduction. The second part presents theoretical and empirical arguments regarding the relation between geographical indication and territorial development. Next, the methodology is described followed by the data analysis with a thorough investigation of the effect of geographical indications on some variables, which are representative of territorial development. The last sections include final considerations and the references used in this study.

2. Geographical indications and their relation to Territorial Development

The purpose of recognizing geographical indications was (and is) a protective commercial practice (Barjolle, Sylvander, & Thévenod-Mottet, 2011; Sautier, Biénabe, & Cerdan, 2011); however, even if they were not designed aiming at development or preservation of the cultural and natural heritage, they might become instruments with these objectives (Sautier et al., 2011).

A GI might aggregate value to a region which goes beyond monetary value (higher prices, increased income and land value, among others) (Cei, Stefani, Defrancesco, & Lombardi, 2018), since it allows a unique form of symbolic capital linked to origin (Conneely & Mahon, 2015). Geographical indications present a cultural dimension, once the product characteristics and the way it is produced, stored, marketed and even appreciated is inherent to the local community (Belletti & Marescotti, 2011). Aggregating value to a product by preserving its original characteristics allows the remuneration and reproduction of local resources, benefitting not only the productive system, but also the dynamics of the territorial development, the local society and the usually fragile natural resources (FAO, 2010). The collective process might lead to transformations and to development and strengthening of public-private and interinstitutional networks, in both the elaboration of use regulations and the product diffusion and promotion (Cendón & Bruno, 2018). It might also promote a sense of regional pride and cultural identity (Neilson et al., 2018), and include the preservation of specific resources such as land

[2] Cultural identity here is the idea of belonging to a certain social group, which assumes the sharing of cultural traits, costumes, values and beliefs.

use and other environmental and social issues including work conditions (Belletti et al., 2017) and heritage (material and immaterial) (Arfini et al., 2019; Lin & Lian, 2018).

This means, therefore, creating a synergy between the three basic factors of the sustainable territorial development: economic growth (strengthening of the local productive chain and opportunities of diversification and integration of economic activities), environment (biodiversity, landscape, land use, etc.) and the civil society (cohesion of local actors, empowerment, inclusion, etc.) (Belletti & Marescotti, 2011; FAO, 2010). As regards the economic issue, in addition to the increase in product prices and demand, in general, the authors point out the development of other sectors such as tourism, accommodation, restoration, handcraft and other services on the way to multifunctionality. When social aspects are observed, work and collective decision making outstand, along with the establishment and institutionalization of partnerships, increase in human capital with greater qualification and demand for qualified workforce. Finally, environmental aspects are taken into account and better used as a basic condition for the existence of the geographical characteristics that enabled the register of the geographical indication.

Despite discussions related to the development resulting from GIs, studies quantifying the impact of public policies for geographical indications are still scarce and usually focus on one or few variables or a specific case (Cei, Defrancesco, & Stefani, 2018). However, it was possible to find some interesting studies that also employed PSM.

Empirically, Jena et al. (2015) analyzed quantitatively the GIs impact on the wellbeing of the population formed by producers living in the regions with geographical indication for Basmati rice in India and Jasmine rice in Thailand. As a result, when using the Propensity Score Matching method (PSM), they observed that families inserted in the GI area presented a greater annual income increase when compared to families that lived outside that area. They also identified a negative effect of poverty, with reduction in economic deprivation for the families included in the rice GIs.

Sitorus, Harianto, Suharno and Syaukat (2020) evaluated the effect of the Muntok White Pepper GI on productivity, sales price and farmers' income on the Bangka Belitung islands. As a result, when applying the PSM method, they identified a positive impact on productivity and prices as well as the income of those producers that were included in the geographical indications of that product.

Using a similar methodology, Diallo (2017) evaluated the impact of the GI of Comté and Cantal cheese on the milk price in their productive chains and the general value of production in France (between 1995 and 2010). Their results evidenced a null effect regarding price and production value, and no statistically significant differences were found for these variables when milk producers in the GI chains were compared to those that did not belong to that category. Moreover, in addition to the dairy GI, the author also evaluated the Colombian coffee GI, reporting that the GI policy for this product increased the price paid to the producers in over 30%. It seems relevant to point out that in the latter case, the analysis was between countries and the milk investigation considered only producers.

Poetschki (2018) analyzed the importance of wine and olive GIs in the farm income in the European Union between 2014 and 2015. Using the endogenous commutation regression models and PSM, that author identified that, on average, producers with olive GI had a higher net income than their colleagues that were not included in the GI, even if the difference was not significant. Regarding wine producers, those without the geographical indication obtained a significantly higher income, which might have resulted from the increase in costs that occur with a geographical indication.

The consideration of previous theoretical and empirical studies determined the methodological choices for this study, which are presented in the next section.

3. Methodological elements

This paper aimed to analyze the effect of geographical indications (GI) on the process of territorial development in Brazilian municipalities, focusing on coffee and wine GIs. The Propensity Score Matching (PSM) method was employed with the objective of evaluating the GIs scope in employment generation, tourism employment generation, exports, rural exodus, economic growth and

socioeconomic development. These variables were chosen due to the availability of data for all Brazilian municipalities, as well as their recurrent mention in the literature. PSM was chosen in detriment of, for example, the difference in difference (DD or DiD), since the GIs of each product were created in different periods, and, therefore, it was impossible to stack the different municipalities versus their characteristics before and after the GI. For this reason, PSM seemed to be the best choice.

Next, we present the econometric method that was used along with the empirical strategy employed.

3.1 Econometric Model: Propensity Score Matching

An efficient way of evaluating the effect of a public policy on certain variable is to observe it when the analysis unit **i** (in this case, municipalities) benefits from certain policy *versus* when it does not enjoy such benefit (Heinrich, Maffioli, & Vázquez, 2010). The difference obtained for the variable of interest consists in the impact of the public policy. That is:

$$\delta_{\mathrm{i}} = \Upsilon_{1\mathrm{i}} - \Upsilon_{0\mathrm{i}} \tag{1}$$

Where: $\boldsymbol{\delta_i}$ is the effect of the public policy – in this case, the GIs – on the variable of interest (*proxies* of territorial development), in relation to the analysis unit **i** (municipalities); $\boldsymbol{\Upsilon_{1i}}$ is the value of the variable of interest after the municipality **i** takes part in the GI; and $\boldsymbol{\Upsilon_{0i}}$ denotes the value of the variable of interest if municipality **i** were not included in the GI.

The impact of any public policy should not be individualized, but rather have its mean effect on the group under analysis measured. For that, the parameter known as Average Treatment Effect (ATE), described as (2) is used.

$$\mathrm{ATE} = \mathrm{E}(\delta) = \mathrm{E}(\Upsilon_1 - \Upsilon_0) \tag{2}$$

Where: **E** corresponds to the expected value; $\boldsymbol{\Upsilon_1}$ is the variable of interest value after the municipality participation in the GI; and $\boldsymbol{\Upsilon_0}$ is the variable of interest value if the municipality does not take part in the GI.

The difficulty of measuring (2) is that this and other effects are not necessarily observable. Thus, assuming that the difference between the means is given by the mean of differences, then ATE can be represented by (3).

$$\mathrm{ATE} = \mathrm{E}(\Upsilon_1|\mathrm{T}=1) - \mathrm{E}(\Upsilon_0|\mathrm{T}=1) \tag{3}$$

Where: $E(Y_0|T=1)$ is the mean results that the treated municipalities would have obtained in the absence of treatment, an element that is not observed. Therefore, this value is substituted with $E(Y_0|T=0)$, measuring the variable of interest value, Y_0, of the group of municipalities that is not benefitted by the GI, which is observed. From that, we can calculate:

$$\Delta = \mathrm{E}(\Upsilon_1|\mathrm{T}=1) - \mathrm{E}(\Upsilon_0|\mathrm{T}=0) \tag{4}$$

If adding and subtracting from (4) the term $E(Y_0|T=1)$, the difference between Δ and ATE (5) is obtained:

$$\Delta = \mathrm{E}(\Upsilon_1|\mathrm{T}=1) - \mathrm{E}(\Upsilon_0|\mathrm{T}=1) + \mathrm{E}(\Upsilon_0|\mathrm{T}=1) - \mathrm{E}(\Upsilon_0|\mathrm{T}=0) \tag{5a}$$

$$\Delta = \mathrm{ATE} + \mathrm{E}(\Upsilon_0|\mathrm{T}=1) - \mathrm{E}(\Upsilon_0|\mathrm{T}=0) \tag{5b}$$

$$\Delta = \mathrm{ATE} + \mathrm{SB} \tag{5c}$$

Where: **SB** corresponds to the difference between the variable **Y** of the group of municipalities benefitted by the GI and those that are not benefitted. If **SB** is equal to zero, the **ATE** can be measured

through the difference between the variable **Y** means; however, this hardly ever occurs, and, therefore, a suitable methodology is required to guarantee that the term **SB** is equal to zero.

One of the methods used to minimize the selection bias, mainly when the selection of the participants in a certain program is not randomized[3], corresponds to the Propensity Score Matching (PSM). Its use requires two groups: the treatment and the control. The first comprises the municipalities that received geographical indication, while the second gathers those municipalities that do not have this indication.

The difficulty found to measure the GI effects between these two groups ends with the distinction of the existing characteristics between both. These characteristics might determine the difference in the variable of interest. In fact, Cavalcanti et al. (2016) inferred that the ideal evaluation of the impact of any public policy should be the comparison between the same group within two contexts: taking part or not in a policy. However, since this is not possible, the alternative lies in the creation of a statistically identical group, so that the differential is only supported by the fact of taking part or not in a specific public policy.

The PSM method allows such comparison, finding the similarities within the groups of municipalities that received the geographical indication (treatment group, T=1) and those that did not receive it (control group, T=0). Such propensity is generated from the observable characteristics of the municipalities, which affect their likelihood of having a GI. Thus, the participation or not of the municipalities in the GI gains a random character (Heinrich et al., 2010). This matching can be carried out in different ways: the closest neighbor method; closer neighbors without substitution; caliper matching; and Kernel. Since these are different environments, we opted for the most used method in this literature, which is the closest neighbor (three neighbors, with repositioning) and the 0.01[4] caliper.

It seems relevant to mention that when seeking similarities when different characteristics are listed, one might incur in the problem of multidimensionality, that is, municipality **i** might show some characteristics that are close to those of municipality **j** and others to **m**. In such case, the difficulty would be to establish with whom **i** should be compared. As pointed out by Rosenbaum e Rubin (1983), the PSM method minimizes this problem by calculating the likelihood of a municipality ending up receiving the treatment, taking into consideration its observable characteristics (covariate). In this research, this likelihood was measured by the Logit model (6), with matching based on the assumption of conditional independence. As a consequence of this assumption, the difference between the covariate means of the treatment and control groups cannot be statistically different from zero.

$$P(T_i = 1|X_i) = \frac{1}{1 + e^{-X_i'\beta}} \quad (6)$$

Where: $P(T_i=1|X_i)$ denotes the likelihood of a municipality taking part in the GI, considering its covariates.

Next, the PSM method carries out the matching between the municipalities with the same or close scores and, then, evaluates whether the treatment group obtained Jobs, Jobs in the tourism sector, Gross Domestic Product per capita, exports, rural exodus and Firjan Municipal Development Index (IFDM) that are statistically higher than those of the control group.

[3] Due to the non-randomization, it is impossible to separate the GI effects from the effects of other factors that might also influence the variable of interest.

[4] In the case of the closest neighbor with repositioning, it is based on the propensity scores, in which for each participating municipality **i** the closest non-participant municipality **j** is selected (one-to-one matching). For the caliper, the closest neighbor with a pre-defined distance of a caliper is sought, so that for each participating municipality **i** there is a non-participant municipality **j** with an estimated propensity score that is within the caliper chosen.

3.2 Empirical strategy

To use the PSM method, it is necessary to have a policy or program to be evaluated, variables of results to be analyzed and the covariates, which are used to compare the results between similar pairs.

In the Brazilian context, GIs are a complex and heterogeneous phenomenon (in relation to products, types of actors, strategies and support networks that enable the register) with a component of broader movements of support and promotion of handcraft products (Wilkinson, Cerdan, & Dorigon, 2017). Due to this heterogeneity, the policy investigated in this paper refers to geographical indications (GI), focusing on the following products: wine and coffee.

According to Belletti et al. (2017) "wine and coffee are two important and iconic examples of markets where the origin plays a significant role. Both products derive from perennial crops that involve the producers in long term relationships with the place and the local resources that have great impact on the soil use and on the landscape".

The result variables include aspects of territorial development, namely, **Firjan Municipal Development Index (IFDM)**, proxy of the population wellbeing; **Gross Domestic Income *per capita***, measuring economic growth; **formal employment per thousand inhabitants**, as a proxy for the labor market; **exports**, measuring the external insertion of the municipality; **employment in the tourism sector per thousand inhabitants**, measuring tourism development, and; **rural exodus**, considering the hypothesis that the geographical indication might improve, in terms of demand and price, the income in the rural area and, consequently, reduce the migration of the rural population to urban centers. These outcome variables were chosen considering other studies that also investigated the impact of GIs on local development[5].

Chart 1 shows the source of these variables, as well as their formulation, emphasizing that the year of collection was the most recent existing information at the municipal level.

Chart 1. Sources of result variables and their composition

Proxy	Variable	Composition	Year	Source
Economic development	Firjan Index	Firjan Index (education, income and health dimensions)	2016	FIRJAN
Economic growth	Gross Domestic Product	Sum of all final goods and services	2017	IBGE
Formal income	Formal employment per thousand inhabitants	$\left(\frac{\text{Total formal employment}}{\text{Estimated population}}\right)1000$	2018	RAIS and IPEADATA
Development of the tourism sector	Formal employment in tourism per thousand inhabitants	$\left(\frac{\text{Tourism employment}}{\text{Total formal employment}}\right)1000$	2018	RAIS
International trade insertion	Exports	Exports of GI products (coffee and wine)	2019	MDIC
Rural exodus	Rural population variation	Rural population 2017 – rural population 2006	2017-2006	IBGE

To choose the covariates, the international literature was surveyed for the variables usually used by authors with similar studies. However, most studies found investigated the GI effects on individuals (such as farms, producers, etc.), and this research presented an aggregated analysis, that is, the

[5] In the work of Jena et al (2015) there is an analysis of the impact of the GI on the quality of life of the population and also on exports. Poetschki (2018) and Diallo (2017) focused on the effect of GIs on the economic environment, whether on property, as well as on the region in which the GI was inserted (focusing on income, total product, employment, etc.). In other works, as in Mendonça et. al (2019), the relationship of GI with rural development and with tourism routes was investigated.

analysis focused the municipal territorial development. Therefore, it was based on the covariates used by authors such as Jena et al., 2015 e Sitorus et al. (2020), which were adapted for a municipal coverage, as shown in Chart 2, highlighting that IBGE was the source of this data.

It seems relevant to emphasize that the Appendix shows the correlation between the covariates of each GI, highlighting that in neither of the cases the value was over 0.80, which according to Oliveira (2016), would be the maximum correlation so as not to distort the PSM results.

Moreover, as a balance test, the test of averages and the Kolmogorov-Smirnov distribution test were employed, whose null hypothesis is that the treatment and the control groups are statistically equal. It seems relevant to highlight the need for the discrepancy of the covariates selected between the groups before the matching. Therefore, those variables that did not meet this requirement in each GI analyzed were not included in the matching.

Chart 2. Covariates used for each GI group

GI	
WINE	COFFEE
Wine production	Coffee production
Producers' mean age	Producers' mean age
Business mean size	Business mean size
Percentage of producers that completed elementary school at most	Percentage of producers that completed elementary school at most
Percentage of family farming	Percentage of family farming
Participation of the farming Gross Domestic Product (GDP) in the total GDP	Participation of the farming Gross Domestic Product (GDP) in the total GDP
Number of grape-producing farms	Number of coffee-producing farms
Grape production	-

Also, regarding the estimates carried out, due to the satisfactory size of the data base used, it was possible to use more restrictive options in relation to the formation of the control group, highlighting the use of the common support criterion and the imposition of trimming in 10% of the observations of the treatment group, which coincided with the smaller densities of the propensity score of the control.

4. GIs and Territorial Development: Empirical Evidence for Brazilian Municipalities

Geographical indications can affect directly the development of certain region. When an origin seal is obtained, a quality logo is obtained, minimizing the transaction costs for the consumer, creating an incentive for them to pay a prize price for the good originated in a certain region with GI, or intensify the demand for these products (Jena et al., 2015). Moreover, when marking successfully the geographical frontier of the production of a certain good, the GI protection controls effectively the offer, which, in turn, increases its price. In addition, the GI protection preserves the traditional heritage of the local that produces a good GI with positive externalities in both the short and long terms.

In this sense, theoretically, a positive effect of a GI is expected on territorial development, with direct impact on income, employment and the population general wellbeing. Next, we analyze this effect, verifying whether in fact there are benefits in receiving a GI, focusing on the geographical indications of wine and coffee. Chart 3 presents the geographical indications of these products already recognized in the country, their geographical name, category (indication of provenance – IP or denomination of origin – DO), year of recognition, federative unit where this area is located and number of municipalities that it covers (totally or partially).

Chart 3. Description of GIs belonging to the analysis groups

Product	Geographical name	Type	Year	FU	Nº of municipalities
Wine	Vale dos Vinhedos	IP	2002	RS	03
	Pinto Bandeira	IP	2010	RS	02
	Vale da Uva Goethe	IP	2012	SC	08
	Vale dos Vinhedos	DO	2012	RS	03
	Altos Montes	IP	2012	RS	02
	Monte Belo	IP	2013	RS	03
	Farroupilha	IP	2015	RS	01
	Campanha Gaúcha	IP	2020	RS	14
Coffee	Região do Cerrado Mineiro	IP	2005	MG	55
	Norte Pioneiro do Paraná	IP	2012	PR	45
	Alta Mogiana	IP	2013	SP	15
	Cerrado Mineiro	DO	2013	MG	55
	Região de Pinhal	IP	2016	SP	08
	Oeste da Bahia	IP	2019	BA	11
	Mantiqueira de Minas Gerais	IP→ DO	2011 → 2020	MG	25

Source: INPI, 2020

Next, we address each of the sectors regarding the results of the GIs under analysis.

4.1 Wine GI

Despite some geographical names being used to refer to products (wine, oil, cheese, bread, olive, among others) since ancient times when people noticed that the products of a certain place presented particular qualities (Faria, Oliveira, & Santos, 2012; Mendes & Antoniazzi, 2012), the formalization of the GI use and the need for legal protection started with wine (Belletti et al., 2017; Bruch, 2011). The Port Wine region (Portugal) is considered to be the first to receive state protection, an action promoted by the Marques de Pombal.

In Brazil, the first geographical indication recognized was also for the same product in the Vale dos Vinhedos (RS) in 2002. The start of its project preceded the approval of the Law Nº 9. 279/1996, resulting from the need to protect it from the competition with the wine that came from the old and new world, mainly within Mercosul, and build up a quality product to compete in the global market, its initiative was pioneer and helped to shape the legal mark existing in the country (Niederle, 2011). It was also in that region that the first case of two GI species with the same geographical name occurred, but it seems relevant to mention that the areas and requirements for the use of the IP and DO are different.

Currently, there are eight wine GIs (from different categories [still, semi-sparkling and sparking; red, white, rose; fine and table; etc.], which are made of several types of grapes): Vale dos Vinhedos (IP and DO); Pinto Bandeira (IP); Vale das Uvas Goethe (IP); Altos Montes (IP); Monte Belo (IP); Farroupilha (IP) and Campanha Gaúcha (IP). There is great geographical proximity between some of them, and some municipalities in Rio Grande do Sul presente part of their territory in one GI delimitation and part in another. The INPI has received applications for recognition of IP for Santa Catarina wine and other three producers are known to be in phase of structuring (Vale do São Francisco; Altos de Pinto Bandeira; and Região do Planalto Catarinense) (Embrapa Uva e Vinho, n.d.).

In fact, currently there are thirty-six municipalities with delimitated areas for wine geographical indications, all in southern Brazil. Figure 1 shows that most wine GIs are close one to another, mainly located in Rio Grande do Sul. These municipalities included in the geographical indications are

responsible for 38% of the grape produced in the country, responding for 24% of all wine produced. At the same time, their insertion in the international trade is quite high, so that, from all the wine exported by the country, 84% came from municipalities that held wine GIs (Table 1). These exports might result from the geographical indication, with the valorization of the product at the international level.

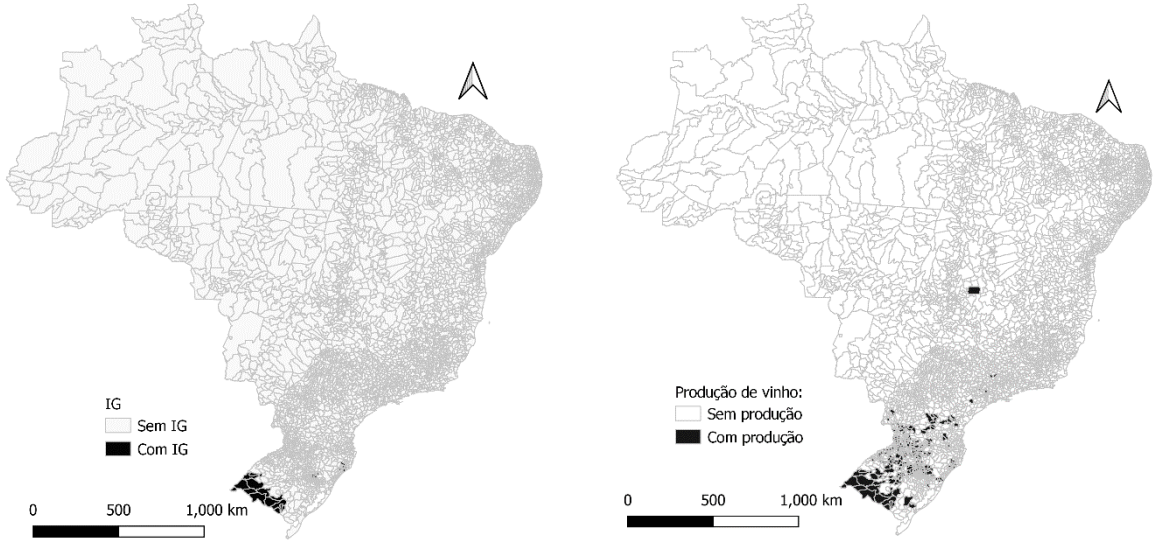
Table 1. Percentage of grape and wine production and wine exports – with and without wine GIs– Brazilian municipalities

	Grape production -2017	Wine production - 2017	Wine exports – 2019
With GIs	38.2	24.3	83.9
Without GIs	61.8	75.7	16.1

Source: IBGE and Comexstat.

Figure 1. Wine GI (A), wine production (B) – Brazilian municipalities

(A) (B)



Source: INPI and IBGE

From these initial considerations, we analyzed the effect of having a Wine Geographical Indication on some of the variables selected related to the territorial development. Therefore, initially we verified how the covariates (characteristic of each municipality) interfered in the probability of having a wine GI (Table 2).

As a result, municipalities with greater grape production and/or a more expressive number of vineyards, intensified their chances of having wine geographical indications. In fact, it is very difficult for a place to achieve this indication if they do not have the local production of the raw material needed. Likewise, to have a wine geographical indication, it is necessary to produce it, therefore, the greater this production is, the higher the chance of getting a geographical indication is.

In addition, municipalities that presented greater participation in the farming GDP increased their chances of having a wine GI. Perhaps, since the economic dynamics is mainly based on the primary sector, instruments (for example, policies) might be used aiming to achieve this indication more easily than in municipalities in which other sectors are more relevant.

Finally, family farming was seen to be important in this process, however, an inverse relation is observed so that places with high percentage of small producers usually have lower changes of obtaining wine geographical indications. Dullius (2009) observed that despite the GI theme having great affinity with the notion of family farming, in Brazil it is also marked by a movement of creation of a differential for agrobusiness products. In addition, Belas (2012) pointed out that specific programs for the commercial valorization of family farming products have been neglected in the discussions

about GIs, which have prioritized the certification brands, mainly the seal of identification for family farming products.

Table 2. Results of the logit model – likelihood of taking part in wine GIs– Brazilian municipalities

Variables	Coefficient	Standard error
Grape production	0.0001*	0.00001
Number of business producing grapes	0.002*	0.001
Wine production	0.001*	0.0001
Percentage of producers that completed elementary school at most	0.47	1.69
Mean age of the producers responsible for the businesses	0.06	0.07
Mean size of businesses	0.0001	0.0001
Percentage of family farming	-0.40*	0.01
Participation in the farming GDP	0.02*	0.01
Pseudo R	0.35	

Note: * Significant at 5%

Finally, family farming was seen to be important in this process, however, an inverse relation is observed so that places with high percentage of small producers usually have lower changes of obtaining wine geographical indications. Dullius (2009) observed that despite the GI theme having great affinity with the notion of family farming, in Brazil it is also marked by a movement of creation of a differential for agrobusiness products. In addition, Belas (2012) pointed out that specific programs for the commercial valorization of family farming products have been neglected in the discussions about GIs, which have prioritized the certification brands, mainly the seal of identification for family farming products.

After analyzing the likelihood of a municipality taking part in the Wine Geographical Indication, aiming at the correct application of the PSM method, the set of Brazilian municipalities was divided into treated (those with wine GI) and control (those without wine GI), verifying whether the characteristics of both groups were uneven before matching[6] and similar after the propensity (Table 3). As a result, this requirement was confirmed, with the possibility of making direct comparisons of these groups in the evaluation of the wine GI effect on territorial development.

Table 4 shows the mean effects of the wine geographical indications on some selected variables, which represent territorial development.

One of the main arguments put forward by those who defend geographical indications refers to the intensification of the international insertion of these products, resulting from the specific quality (or notoriety) associated to the geographical origin. This is verified in the case of wines, so that having a geographical indication, on average, increased municipal exports in approximately US$ 276 thousand/year. Some articles, such as that of Linder (1961) and Crozet et al. (2012), infer that higher quality products tend to have a greater international insertion, receiving higher prices. The problem is that the "quality" of a product is an unobservable attribute, difficult to measure. Theoretically, with the geographical indication, this informational asymmetry between consumers and producers is mitigated, presenting itself as a "seal" that attests to the reputation of the offered products (Agostino & Trivieri, 2014; Raju & Tiwari, 2015; Ribeiro, 2020). This is verified in the case of Brazilian wines, justifying their greater competitiveness in the international market.

[6] In the Kolmogorov-Smirnov distribution test, Ha (statistical value equal to 2.14) was rejected, indicating that before the matching there was difference between the groups (treated and control).

Table 3. Difference of means between the covariate variables before and after matching – Wine geographical indications – Brazil

Variables	Before treatment			After treatment		
	Untreated	Treated	t test p-value	Untreated	Treated	t test – p-value
Grape production	129	15150	0.00*	9340	15151	0.42
N° of business producing grape	19	182	0.00*	189	182	0.95
Wine production	1.81	107	10.6*	101	107	0.66
Mean age of producers responsible for the businesses	51	52	0.91	51	52	0.75
Percentage of producers that completed elementary school at most	0.85	0.81	0.05*	0.79	0.81	0.44
Percentage of family farming	73	64	0.00*	57	65	0.22
Mean size of businesses	94	176	0.99	366	176	0.20
Participation in the farming GDP	18	25	0.05*	29	24	0.27

Note: * significant at 5%

Indirectly, this result might be linked to the maintenance of the rural population in the countryside, mainly for improving the local income, by adding value to rural products (Mendonça et al., 2019). In fact, when analyzing the wine GI impact on rural exodus, a negative and statistically significant value was obtained. That is, approximately 952 more people remained in the rural zone in the municipalities with GI when compared to those that did not have this indication.

Table 4. Mean Effect of Wine Geographical Indications on the selected variables

Result variable	Method	Mean effect	Std. Err.
Firjan Index	Caliper	**0.08***	0.03
	Neighbor	**0.09***	0.03
Rural Exodus (variation)	Caliper	**-952.7***	360.23
	Neighbor	**-580.40***	268.63
Wine exports (US$)	Caliper	**276488****	167177
	Neighbor	**275864****	166779
GDP per thousand inhabitants (R$)	Caliper	6.03	6.01
	Neighbor	5.69	4.35
Employment per thousand inhabitants	Caliper	**82.6***	25.99
	Neighbor	**82.3***	21.3
Employment in tourism per thousand inhabitants	Caliper	**0.18***	0.08
	Neighbor	**0.19***	0.08

Note:* level of significance 5%.

As our results indicate a direct contribution of wine GIs to rural development and, simultaneously, there was a greater chance of occurrence of wine GIs in regions with fewer family establishments (Table 2), attention is needed to public policies that promote origin products. Such policies focused in enhancing agricultural products value could prioritize the relationship between family farming and

existent and potential geographical indications throughout Brazil, aiming to make the impacts of the IG even more expressive.

Moreover, the job market was seen to be heated in the municipalities with GI, with the creation of an excess of 82 employments per thousand inhabitants. This generation does not result from jobs in the wine production area only, but indirectly, from the higher local income, with improvement of the job multiplier. Discussions related to GIs have shown the potential effect of increased tourism (Čehić, Mesić, & Oplanić, 2020; De La Torre, Fernández, & Pérez Naranjo, 2014; Jiménez, de la Torre, & Rojas, 2019; Pasli & Akbaba, 2019). This data was confirmed in this study, since the tourism sector benefitted from the wine geographical indications, showing some heating in this sector labor market, exceeding in around 18 jobs per thousand inhabitants in the municipalities with GI. This number can be even higher since some organizations hire enologists to work on both the wine production and the reception of tourists and other activities related to the wine retailer market and this job, therefore, is not necessarily recorded as tourism-related employment (Falcade, 2005; Tonini & Lavandoski, 2011).

At the same time, no statistical difference was observed regarding economic growth. Therefore, if on the one hand the GI speeds the formation of work positions, on the other hand, it does not overlap the municipality in terms of production of final goods and services.

This does not mean that the population wellbeing is not raised, on the contrary, we found a positive effect of the wine GIs on the Firjan Municipal Development Index. Some authors such as Sant'Ana (2008), inferred that obtaining economic growth does not necessarily mean obtaining development. Mainly in developing countries, a high concentration of production factors is observed that make the economic growth obtained to be concentrated on few agents, which does not promote wellbeing. In the case of wine geographical indications, since they include mainly small producers, the results indicate better distribution, with improvements of wellbeing for the local population.

Therefore, the wine geographical indications were seen to affect positively the main variables selected for territorial development, appearing as important factors in this process. Currently, there are 32 municipalities with formal application for wine GIs with the INPI (in addition to others that are included in the GI structuring phase), and these results show the importance of such achievement aiming at the improvement of local aspects of the wine producing regions.

4.2 Coffee GI

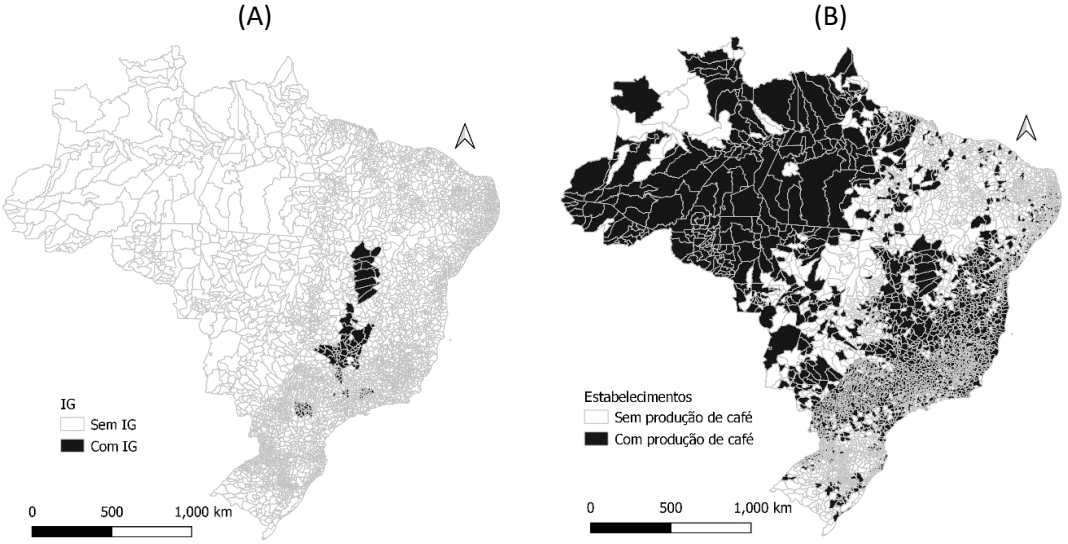
Since the XIX century, coffee has had great importance in the Brazilian economy for being one of its main export products (Giesbrecht, Minas, & Gonçalves, 2014). It can also be considered one of the products in the food industry that is most consumed worldwide (Lourenzani, Watanabe, Pigatto, & de Godoi Pereira, 2019). Three "waves" are considered to influence the global coffee market and consumption. The most recent (third wave) is marked by a radical change in the product perception (currently considered as complex as wine) (Guimarães, Castro Júnior, & Andrade, 2016) and an increase in the demand for differentiated quality products valorization of specific origins and its *terroir*, valorization of a sensorial and singular experience of consumption and greater sensitiveness regarding environmental sustainability and the challenges and vulnerabilities faced by coffee producers (Guimarães et al., 2016; Lourenzani et al., 2019). In this scenery, the recognition of geographical indications might become interesting.

In the Brazilian case, the first geographical indication recognized for this sector (and second in general) was the indication of origin conferred to the Região do Cerrado Mineiro in (2005). Currently, there are 7 GIs in this sector, from which 5 are IP (Região do Cerrado Mineiro; Norte Pioneiro do Paraná; Alta Mogiana; Região do Pinhal and Oeste da Bahia) and 2 are DO (Cerrado Mineiro and Mantiqueira de Minas Gerais). Regarding Mantiqueira, the IP was firstly obtained in (2011) whose typology was altered in 2020. This alteration had not been possible before (only after the Normative Instruction INPI nº95/2018), for this reason, the IP and the DO coexist in the Cerrado Mineiro region. In addition to the already recognized GIs, there are at least 5 pending applications of other Brazilian

coffee brands being analyzed at the INPI (Matas de Minas, Caparaó, Espírito Santo, Montanhas do Espirito Santo, and Matas de Rondônia) which comprise (total or partially) 157 Brazilian municipalities.

Inserted in already delimited and recognized areas, currently, there are a hundred and fifty-nine municipalities included in the coffee geographical indications, located in the South, Southeast and Center-West regions of Brazil. Figure 2 shows that most of the coffee GIs are close one from another, mainly located in the Southeast of the country, with 65% of all municipalities included. These municipalities that have coffee GIs are responsible for 21% of the production in the country, however, holding only 11% of the Brazilian coffee exports (Table 5). As a whole, from all municipalities that produce coffee, 12% have a geographical indication.

Figure 2. Coffee GIs (A) and number of coffee-producing businesses (B) – Brazilian municipalities - 2017



Source: INPI and IBGE

Table 5. Percentage of coffee production and exports – with and without GIs – Brazilian municipalities

	Coffee production -2017	Coffee exports – 2019
With GIs	20.8	10.8
Without GIs	79.2	89.2

Source: IBGE and Comexstat.

In the analysis of the coffee GI effect on territorial development, it is interesting to firstly verify how the covariates interfere in the likelihood of having a coffee GI (Table 6). The results demonstrate that having a greater coffee production and/or having a high percentage of coffee-producing businesses has a positive impact on the chances of a municipality having a GI. In addition, greater participation of the farming GDP in the economy and older age of those responsible by the properties, on average, intensify the likelihood of obtaining a geographical indication. Finally, the producers' level of schooling is a relevant element in this process.

After analyzing the likelihood of taking part in the coffee geographical indication, the verification of whether the characteristics of the treated municipalities (with GI) and those of the control municipalities (without GI) were different before matching[7] and similar after the propensity (Table 7). By using the mean difference test, the requirement of the PSM method was confirmed, so that the characteristics selected (covariate) became statistically equal after matching.

[7] In the Kolmogorov-Smirnov distribution test, Ha (statistical value equal to 1.94) was rejected, indicating that there was difference between the groups (treated and control) before matching.

Table 6. Results of the logit model – likelihood of obtaining a coffee GI – Brazilian municipalities

Variables	Coefficient	Standard error
Coffe production	0.0002*	0.00003
Number of businesses that produce coffee	0.001*	0.0005
Percentage of family farming	0.01	0.01
Mean age of producers responsible for the business	0.09*	0.03
Percentage of producers that completed elementary school at most	-4.27*	0.75
Mean size of businesses	-0.001	0.001
Participation in the farming GDP	0.03*	0.005
Pseudo R	0.33	

Note: * Significant at 5%

Table 7. Mean difference between covariate variables before and after matching– Coffee geographical indication – Brazil

Variables	Before treatment			After treatment		
	untreated	Treated	t test p- value	untreated	Treated	t test p-value
Coffee production	364	3258	0.0*	2066	3258	0.14
Coffee-producing businesses	53	122	0.01*	90	122	0.21
Mean age of producers responsible for the business	50	52	0.00*	53	52	0.15
Percentage of producers that completed elementary school at most.	0,85	0.76	0.00*	0.77	0.75	0.22
Percentage of Family farming	72	68	0.00*	68	68	0.75
Mean size of business	95	106	0.76	112	106	0.72
Participation in the farming GDP	18	27	0.00*	30	27	0.15

Note: * Significant at 5%

Table 8 shows the mean effects of the coffee geographical indications on some selected variables, which represent territorial development. The theoretical expectation was that a differentiated impact on exports would be observed in the municipalities with coffee GI; however, the difference found was not statistically significant. This might result from the low representativeness of coffee with GI exports in relation to the total that is traded abroad, which corresponds to only 11% (Table 5).

The literature reports that, in addition to the market growth resulting from the decrease in the information asymmetry to the consumers, the geographical indication also provides a prize in terms of product price. Diallo (2017), when analyzing the international insertion of coffee with geographical indication in Colombia, observed a 30% difference in the price paid to the produces. Therefore, we tested the existence of this prize for the Brazilian producers, and identified and increase of around USD 0.77 per kilogram of coffee exported. Thus, although there is no differentiation in terms of the amount exported by the Brazilian producers with GI, these exporters receive a prize, which values the product internationally.

This higher income impacts both the municipal economic growth and the employment generation. As for the former, having a coffee geographical indication generated a BRL 3 thousand additional GDP and, regarding the latter, 31 more job positions were observed to be created in the municipalities with GI when compared to the other municipalities.

This higher dynamics also affected positively the Firjan Municipal Development Index, representing an additional in terms of the population wellbeing in those municipalities with coffee GI.

Table 8. Mean effect of the coffee geographical indications on the selected variables

Result variable	Method	Mean effect	Std. Err.
Firjan Index	Caliper	**0.03***	0.009
	Neighbor	**0.03***	0.009
Rural Exodus (variation)	Caliper	74.98	167.55
	Neighbor	142.74	115.80
Coffee exports (USD)	Caliper	2513421	2894452
	Neighbor	2866133	2011214
Mean price received by coffee exports (USD/kg)	Caliper	**0.77***	0.31
	Neighbor	**0.78***	0.31
GDP per thousand inhabitants (BRL)	Caliper	**3.05****	1.78
	Neighbor	**3.68***	1.37
Employment per thousand inhabitants	Caliper	**31.26***	9.30
	Neighbor	**30.64***	7.45
Tourism employment per thousand inhabitants	Caliper	**0.04****	0.02
	Neighbor	**0.02****	0.01

* Significant at 5%; ** * Significant at 10%

Finally, an extra generation of employment in the tourism sector was also identified in the municipalities with coffee geographical indication. Quality tourism can only be achieved through the maintenance of the natural environment and the existing cultural heritage. In this sense, the geographical indication is decisive for the safety and protection of the geographical name, allowing the delimitation of the area with valorization of the local products and/or services (Bianchini et al., 2016). Thus, theoretically, a relation between geographical indications and the tourism sector dynamics was expected. Regarding the coffee GIs, this phenomenon was confirmed, presenting the indications as an instrument for the local tourism development policy.

5. Conclusions

Geographical indications can be understood as a collective asset that enables the protection and valorization of products through the articulation of actors and their strengthening for the production of goods (or services) that are registered, as well as of other activities, contributing to the territorial development. Thus, this study sought to evaluate the effect of geographical indications (GIs) on the territorial development process in Brazilian municipalities, focusing on existing coffee and wine GIs, through a quantitative analysis using the Propensity Score Matching.

In this method, a comparison is carried out between the treatment group (in this case the municipalities that have a recognized geographical indication) and the control group (those that do not have a GI). The control group, however, comprises municipalities that are statistically similar in relation to some characteristics and that, theoretically, would also be able to apply for a GI. The analysis of covariate variables for the proper matching needed in the PSM already produces some relevant information regarding the structuration process for the application. Both sectors under study presented significance for raw-material production (*in natura* grapes and coffee) and the final product (wine and processed coffee), which had been expected, since having the production predominantly within the delimited area is a basic requirement to obtain this product/service GI. Also, having greater participation in the farming GDP of the municipality was seen to favor the creation of policies and actions for this sector, as can be seen in the case of geographical indications.

Other covariate variables had different significance according to the productive chain: I) older age of those responsible for the business, as well as higher schooling level impacted positively on the chance of obtaining coffee GIs; II) The business size had an opposite significant impact in the case of wine GIs, presenting a positive relation, indicating that larger properties represent higher chances of obtaining a GI; III) Higher number of family farms impacted inversely the wine GI, showing that the higher the number of small producers is, the lower the chances of obtaining a GI is.

Regarding the main results after matching, the group that had GI obtained better indicators in most of the variables selected to measure territorial development. More specifically, the increase in the number of formal jobs was found in the two GI groups under study, indicating that more positions are found in the job market in municipalities where there are recognized GIs. This might occur with jobs directly related to the relevant production, or other areas such a tourism, events, logistics, etc. In the specific case of formal jobs in the tourism sector, an increase in the job positions in the coffee and wine GIs was observed, and the latter presented enotourism as a consolidated and increasing segment in part of the municipalities that had GI in the country.

Positive results were found for the proxies of economic growth (coffee GIs) and population wellbeing (wine and coffee GIs). The economic factor (with increase in the product prices, the land within the GI delimitation, demand increase, client retention, among others) is very recurrent in the literature (Mesić, Božić, & Cerjak, 2017; Raju & Tiwari, 2015; Zhan, Liu, & Yu, 2017), even for the GI commercial origins. Since no data was obtained for all consequential factors, we understood that the gross domestic product per capita could translate the economic growth in general. This fact was verified in the coffee GI.

Economic growth is not necessarily homogeneously distributed among the local actors, therefore, we thought it would be important to use a parameter that represented the population wellbeing. For this reason, we opted for the Firjan Municipal Development Index that considers not only the economic factor, but also the achievements and social transformations based on the official data made available by the Ministries in the areas: Employment & Income, Education and Health. It seems relevant to emphasize that this index has a scale that goes from 0 to 1, and the closer to 1 the score is, the greater the local development is. The results evidenced a positive increase in the Firjan index in the municipalities with wine (0.08) and coffee (0.03) GIs, demonstrating the importance of having a geographical indication when higher levels of socioeconomic development are in the agenda.

The widespread, valorization and protection of export products have been the objective sought in many sectors and countries with GIs. This occurs because the regionalized differentiation of a product might call attention from the international market, since a unique characteristic is offered and the confirmation that the origin stated is exactly that protected by the credibility that generated notoriety. In the Brazilian case under analysis, an increase in the international insertion of the wine GIs was observed. However, for the coffee GIs, no significant increase in the volume of exports was found, but rather an increase in the value was observed (better prices).

Increase in the demand (domestic and/or external) of the product with GI as well as the possible diversification of businesses and consequent increase in employment, income, etc., raises people's interest in residing in the relevant area (Belletti et al., 2017; Cendón & Bruno, 2018; Champredonde & Cosiorovski, 2016). Despite the decrease in rural exodus being expected, it was only verified in the case of wine GIs (in which the mean effect was the permanence of 952 people more than in the municipalities without GI).

The GI discussion as a strategy of development is still beginning in Brazil, many challenges still have to be overcome, mainly those regarding the market, the insertion of family farming in this process, in addition to the understanding about the valorization of products via GIs itself. (Artêncio, Giraldi, & Galina, 2019; Pellin, 2019) Measuring the results of recognized GIs is not a simple task, but it is very important to support decisions made by both public and private managers. Among the factors that hamper this measurement is the lack of data. In the European Union there is a data base with accounting information per rural property (Farm Accountancy Data Network - FADN), which is not

available in our country. In addition, the measurement of more subjective issues (such as pride, belonging, positive image, trust, etc.) required research through contact with the consumer or producer. Therefore, the proxies selected were those publicly available. Using PSM to compile the data from several data bases (IBGE, FIRJAN, RAIS, IPEADATA and MDIC) might be an interesting alternative to longitudinal analysis, as well as to eventual *benchmarking* for future claims within each sector.

Finally, it seems relevant to emphasize that "obtaining a GI register is not such a difficult task, but regulating it and transforming it into a development instrument is a great challenge". The public actors (State, governments and other entities of public interest), as well as supporting public policies might create the necessary conditions to foster the developments linked to their origin in order to favor their positive impacts on economic, social and environmental aspects. However, discussions and empirical studies covering the issues in the short, medium and long term are interesting and desirable in the Brazilian and international contexts. Therefore, more research is still needed. Further studies might gather other consequences listed in the literature; carry out other types of quantitative, qualitative or mixed analyses as well as analyze the likelihood of negative factors occurring according to the characteristics of each region.

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Appendix A: Correlation Between Covariates

Wine GIs

	Qtgrape	Grapebus	farmGDP	Perclem	Family farm%	Size	Age	Qtwine
Qtgrape	1.0000							
Grapebus	0.4399	1.0000						
FarmGDP	-0.013	0.0822	1.0000					
Perclem	0.0174	0.0950	-0.003	1.0000				
Familyfarm%	0.0224	0.1441	0.0351	0.6407	1.0000			
Size	-0.020	-0.074	0.2441	-0.2572	-0.3829	1.0000		
Age	0.0188	-0.009	-0.086	-0.0681	-0.1987	0.0474	1.0000	
Qtwine	0.4276	0.3318	-0.016	0.0088	0.0185	-0.0156	0.0185	1.0000

Note: Qtgrape: refers to the quantity of grape produced; Grapebus: number of grape producing businesses; FarmGDP: Participation of the farming GDP in the total GDP; Perclem: percentage of producers that completed elementary school at most; Familyfarm% refers to the percentage of family farms; Size: mean size of rural properties; Age: mean age of farmers responsible for the businesses; Qtwine: amount of wine produced.

Coffee GIs

	Qtcoffee	Coffeebus	FarmGDP	Perclem	Familyfarm%	Size	Age
Qtcoffee	1.0000						
Coffeebus	0.6566	1.0000					
FarmGDP	0.0084	-0.001	1.0000				
Perclem	-0.069	0.0325	-0.003	1.0000			
Familyfarm%	-0.022	0.0438	0.0351	0.6407	1.0000		
Size	-0.035	-0.067	0.2441	-0.2572	-0.3829	1.0000	
Age	0.0182	-0.018	-0.086	-0.0681	-0.1987	0.0474	1.0000

Note: Qtcoffee: refers to the amount of coffee produced; Coffeebus: number of coffee producing businesses; FarmGDP: Participation of the farming GDP in the total GDP; Perclem: percentage of producers that completed elementary school at most; Familyfarm% refers to the percentage of family farms; Size: mean size of rural properties; Age: mean age of farmers responsible for the businesses.